Lecture 11

Cartan's calculus

Objectives. *Hamiltonian motion moves sets of initial positions and momenta through phase space. We prove that this motion preserves phase-space volume. Pullback compares a form at a moving point with a form at the starting point. Differentiating this comparison gives its Lie derivative. Cartan's formula expresses that derivative using exterior differentiation and insertion of a vector field. We prove the formula, including its time-dependent version, and apply it to volume preservation and Hamiltonian motion.*

By the end of this lecture you will be able to:

- Compute contraction with its first-slot sign convention.
- Prove Cartan's formula and differentiate a time-dependent pullback.
- Determine whether a local flow preserves a chosen volume form.
- Derive Liouville's theorem from closedness and nondegeneracy.

11.1 Phase space and Hamiltonian motion

A particle's position alone does not determine its motion. We also specify its momentum.

Definition 11.1. The *phase space* of a particle moving in $\mathbb{R}^n$ is $T^*\mathbb{R}^n$. A point (q, p) records a position $q \in \mathbb{R}^n$ and a momentum $p \in T_q^*\mathbb{R}^n$.

We write the position coordinates as q^i and the momentum coefficients as p_i. The lower momentum index records its covector transformation law.

Physical input. For a classical Hamiltonian model with smooth energy function $H : T^*\mathbb{R}^n \to \mathbb{R}$, motion obeys Hamilton's equations:

$$\dot{q}^i = \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i = -\frac{\partial H}{\partial q^i}.$$

For a harmonic oscillator of mass $m > 0$ and angular frequency $\nu > 0$, the energy is

$$H(q, p) = \frac{p^2}{2m} + \frac{m\nu^2 q^2}{2}.$$

These equations give $\dot{q} = p/m$ and $\dot{p} = -m\nu^2 q$. Differentiating the first equation gives the oscillator equation $\ddot{q} = -\nu^2 q$. The complete flow is

$$\varphi_t(q, p) = \left(q\cos(\nu t) + \frac{p}{m\nu}\sin(\nu t), -m\nu q\sin(\nu t) + p\cos(\nu t)\right).$$

Differentiation verifies Hamilton's equations and the initial value (q, p). In units with $m = \nu = 1$, the Hamiltonian is $(q^2 + p^2)/2$ and the field is $p\partial_q - q\partial_p$. Its flow becomes

$$\varphi_t(q, p) = (q\cos t + p\sin t, -q\sin t + p\cos t).$$

From $(1, 0)$ it reaches $(0, -1)$ at time $\pi/2$. The figure shows this clockwise motion and the stationary origin.

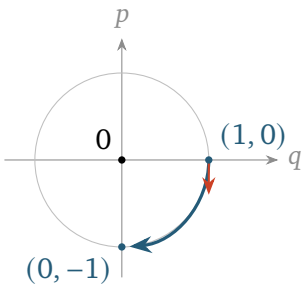


Figure 11.1: Positive-time oscillator motion is clockwise in the (q, p) plane for $m = \nu = 1$. The arc follows the orbit from $(1, 0)$ towards $(0, -1)$, with the straight arrow marking the initial velocity.

We express Hamilton's equations using the forms of Lecture 10.

Definition 11.2. On $T^*\mathbb{R}^n$, the *canonical 1-form* θ evaluates a phase-space tangent vector on its position component using the momentum covector. In these coordinates it is given by

$$\theta = p_i \mathrm{d}q^i.$$

For a position-coordinate change with Jacobian J, write p as a row and $\mathrm{d}q$ as a column. The covector rule of Lecture 10 gives $p' = pJ^{-1}$ and $\mathrm{d}q' = J\mathrm{d}q$. Thus $p'\mathrm{d}q' = p\mathrm{d}q$. The definition is independent of the position coordinates. We denote the following 2-form by ω:

$$\omega = \sum_i \mathrm{d}q^i \wedge \mathrm{d}p_i = -\mathrm{d}\theta.$$

The minus sign follows from $\mathrm{d}\theta = \sum_i \mathrm{d}p_i \wedge \mathrm{d}q^i$. For a tangent field $X = \dot{q}^i \partial_{q^i} + \dot{p}_i \partial_{p_i}$, inserting X into the first argument gives the 1-form

$$\omega(X, \cdot) = \sum_i (\dot{q}^i \mathrm{d}p_i - \dot{p}_i \mathrm{d}q^i).$$

Equating it with $\mathrm{d}H$ recovers Hamilton's equations by comparing coefficients. The form is closed because its coefficients are constant. It is nondegenerate: if this insertion is zero, all coefficients of X vanish. Nondegeneracy lets the energy determine a unique field. We will use closedness to prove preservation of phase-space volume.

11.2 Contraction with a vector field

The insertion used in Hamilton's equations applies to forms of every degree. It also lets us differentiate a pullback without solving for the flow. Let X be a smooth vector field on a smooth manifold M.

Definition 11.3. For $k \geq 1$, the *interior product*, or *contraction*, with X is the map

$$\iota_X \colon \Omega^k(M) \longrightarrow \Omega^{k-1}(M)$$

that fills the first argument. For $p \in M$ and $v_j \in T_pM$ it is given by

$$(\iota_X \alpha)_p(v_1, \ldots, v_{k-1}) = \alpha_p(X_p, v_1, \ldots, v_{k-1}).$$

For a 1-form this is the function $\alpha(X)$. For a function f set $\iota_X f = 0$.

The resulting form is smooth because its coordinate coefficients are sums of products of smooth coefficients of X and α.

Theorem 11.4 (Contraction rules). *Contraction is real-linear, lowers degree by one, and satisfies*

$$\iota_X(\alpha \wedge \beta) = (\iota_X\alpha) \wedge \beta + (-1)^k \alpha \wedge \iota_X\beta, \qquad \iota_X^2 = 0,$$

where α has degree k. The first identity is the graded antiderivation rule. *Contraction is also linear over smooth functions in both X and the form.*

Proof. For 1-forms $\alpha_1, \dots, \alpha_\ell$, expand the determinant defining their wedge along its first column. This gives

$$\iota_X(\alpha_1 \wedge \cdots \wedge \alpha_\ell) = \sum_{j=1}^{\ell} (-1)^{j-1} \alpha_j(X) \alpha_1 \wedge \cdots \wedge \widehat{\alpha_j} \wedge \cdots \wedge \alpha_\ell.$$

The hat means that the factor is omitted. For the wedge of a block of k factors and a second block, split this sum according to the block containing the omitted factor. Omissions in the second block acquire $(-1)^k$. This is the stated graded antiderivation rule. Basis wedges span all forms, so bilinearity extends the identity to arbitrary forms. Applying contraction twice inserts X twice. Alternation makes this zero. On degrees zero and one it is zero by definition. Pointwise multilinearity proves both function-linear assertions and real-linearity. ∎

For example, $\iota_{\partial_x}(\mathrm{d}x \wedge \mathrm{d}y) = \mathrm{d}y$. For the radial field $X = x\partial_x + y\partial_y$,

$$\iota_X(\mathrm{d}x \wedge \mathrm{d}y) = x\mathrm{d}y - y\mathrm{d}x.$$

Filling the last argument instead would reverse this sign. This contraction is the numerator of Lecture 10's angular form.

11.3 Lie derivatives of forms and tensors

Let φ be the maximal local flow of X, supplied by Theorem 9.8. Near each $p \in M$, its maps are defined for all sufficiently small times. Pullback gives a form in the fixed exterior space at p, as in the figure.

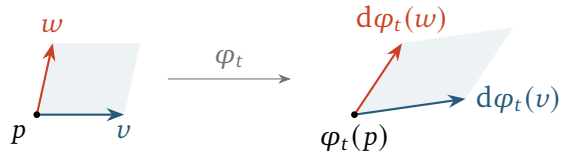


Figure 11.2: Pullback evaluates the form at $\varphi_t(p)$ on the transported vectors $\mathrm{d}\varphi_t(v)$ and $\mathrm{d}\varphi_t(w)$. The pullback at p gives this same evaluation.

Definition 11.5. The *Lie derivative of a form* is the operator

$$L_X \colon \Omega^k(M) \longrightarrow \Omega^k(M)$$

given by

$$(L_X\alpha)_p = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_0 (\varphi_t^*\alpha)_p.$$

In coordinates the pullback has smooth coefficients jointly in time and position. Their time derivatives therefore define a smooth form. For a function, differentiating $f(\varphi_t(p))$ gives $L_X f = Xf$, agreeing with Lecture 9.

Theorem 11.6 (Lie derivative rules · *proved given smooth local flows*)**.** *The Lie derivative is real-linear and preserves degree. It satisfies*

$$L_X(\alpha \wedge \beta) = L_X\alpha \wedge \beta + \alpha \wedge L_X\beta, \qquad L_X \mathrm{d}\alpha = \mathrm{d}L_X\alpha.$$

Proof. Pullback preserves wedge by Theorem 10.14. Differentiating this identity at zero and using the ordinary product rule proves the first identity. No graded sign occurs because the derivative preserves degree. For the second, differentiate $\varphi_t^* \mathrm{d}\alpha = \mathrm{d}(\varphi_t^*\alpha)$. In a fixed chart, exterior differentiation differentiates coefficient functions in space. Time and space derivatives commute by joint smoothness. Thus the time derivative passes through d, proving the assertion. ∎

For example, the field $X = x\partial_x$ has flow $\varphi_t(x) = e^t x$. It gives $\varphi_t^* \mathrm{d}x = e^t \mathrm{d}x$, so $L_X \mathrm{d}x = \mathrm{d}x$.

Remark 11.7 (Tensor pullback)**.** Lie derivatives extend to tensors using the same comparison.

Definition 11.8. For a diffeomorphism $f\colon M \to N$ and a tensor of type (r, s) on N, define f^*T by sending each tangent argument forward by $\mathrm{d}f_p$ and each cotangent argument forward by the dual of $(\mathrm{d}f_p)^{-1}$. These arguments all lie at $f(p)$, where T is evaluated. The *Lie derivative of a tensor* $L_X T$ is the derivative of $\varphi_t^* T$ at zero.

This exists by the same smooth-coordinate argument. For vectors this pullback is $(\varphi_{-t})_* Y$, so it recovers $L_X Y = [X, Y]$ from Lecture 9. General tensor pullback here requires a diffeomorphism. Lecture 10's pullback formula for forms applies verbatim to covariant tensors along any smooth map. For example, on $\mathbb{R}$ the field $X = x\partial_x$ has flow $\varphi_t(x) = e^t x$. The vector field $Y = \partial_x$ pulls back to $e^{-t}\partial_x$. Its derivative at zero is $-\partial_x = [x\partial_x, \partial_x]$. Lecture 16 uses tensor Lie derivatives to characterize flows that preserve a metric.

11.4 Cartan's formula and time dependence

Cartan's formula computes the derivative of a form using only exterior differentiation and contraction. Both are local operations on coefficients.

Theorem 11.9 (Cartan's formula · *proved given smooth local flows*)**.** *On differential forms,*

$$L_X = \mathrm{d}\iota_X + \iota_X \mathrm{d}.$$

Proof. We compare the two operators on functions and their differentials. Write $C_X = \mathrm{d}\iota_X + \iota_X \mathrm{d}$. The two graded product rules imply that C_X is an ordinary derivation. To see the cancellation explicitly, let $\deg \alpha = k$. In $C_X(\alpha \wedge \beta)$ the terms $(\iota_X\alpha) \wedge \mathrm{d}\beta$ have total coefficient $(-1)^{k-1} + (-1)^k = 0$. The terms $\mathrm{d}\alpha \wedge \iota_X\beta$ have coefficient $(-1)^k + (-1)^{k+1} = 0$. The remaining terms are $(C_X\alpha) \wedge \beta + \alpha \wedge C_X\beta$. The calculation also holds when $k = 0$, since contraction of a function is zero.

On functions, $C_X f = \iota_X \mathrm{d}f = Xf = L_X f$. On their differentials, $C_X \mathrm{d}f = \mathrm{d}(Xf) = L_X \mathrm{d}f$, using $\mathrm{d}^2 = 0$ and Theorem 11.6. Both sides depend only on the form near the point in question. For small time the trajectory stays in any chosen chart neighbourhood. Uniqueness makes the restricted field's flow agree with the original flow there. The Lie derivative therefore

commutes with restriction to that chart. Locally every form is a sum of products of functions and coordinate differentials. The derivation rules therefore prove equality on every form in every chart. ∎

This is Tu, Theorem 20.10, with contraction in the first slot. For example, for $X = \partial_x$ and $\alpha = x\mathrm{d}y$, contraction gives zero while $\iota_X \mathrm{d}\alpha = \iota_X(\mathrm{d}x \wedge \mathrm{d}y) = \mathrm{d}y$. Thus $L_X\alpha = \mathrm{d}y$. The flow gives an independent check: $\varphi_t(x, y) = (x + t, y)$, so $\varphi_t^*(x\mathrm{d}y) = (x + t)\mathrm{d}y$ has derivative $\mathrm{d}y$. Cartan's formula acts on forms. It does not replace the vector-field formula $L_X Y = [X, Y]$.

For fixed α, the group law for an autonomous flow also gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\varphi_t^*\alpha = \varphi_t^* L_X \alpha.$$

Indeed, write $\varphi_{t+h}^* = \varphi_t^*\varphi_h^*$ and differentiate in h at zero. This identity holds wherever the indicated local maps exist.

Self-check. For a smooth function $f\colon \mathbb{R} \to \mathbb{R}$ and $X = x\partial_x$, compute $L_X(f\mathrm{d}x)$.

Answer. Cartan gives $\mathrm{d}\iota_X(f\mathrm{d}x) = \mathrm{d}(xf) = (xf' + f)\mathrm{d}x$. The other term is zero because every 2-form on $\mathbb{R}$ vanishes. The flow gives $\varphi_t^*(f\mathrm{d}x) = f(e^t x)e^t\mathrm{d}x$. Differentiating at zero gives the same coefficient $xf' + f$.

11.4.1 Time-dependent transport and fluid volume

A fluid's velocity can change with time. Its trajectories then depend on both their starting time and their current time.

Definition 11.10. A *time-dependent field* V_t has coefficients smooth jointly in $(t, p) \in I \times M$, where $I \subseteq \mathbb{R}$ is an open interval. Its *evolution maps* $\psi_{t,s}$ solve $\partial_t \psi_{t,s}(p) = (V_t)_{\psi_{t,s}(p)}$ with $\psi_{s,s}(p) = p$.

Their local existence and joint smoothness follow by applying Theorem 9.8 to the field $(1, V_t)$ on $I \times M$. Adding time as a coordinate makes this field autonomous. Uniqueness gives the local inverse $\psi_{s,t}$, so each evolution map is a diffeomorphism between its open domain and image.

Definition 11.11. A *smooth family of forms* α_t has coefficients smooth in (t, p). Its derivative $\partial_t \alpha_t$ differentiates those coefficients in fixed, time-independent charts.

The pullback changes because the points move and because the form itself changes.

Proposition 11.12 (Time-dependent pullback · *proved given smooth local evolution maps*). *Fix an initial time $s \in I$ and write $\psi_t = \psi_{t,s}$. On every region where these maps are defined,*

$$\frac{\mathrm{d}}{\mathrm{d}t}\psi_t^*\alpha_t = \psi_t^*\left(L_{V_t}\alpha_t + \partial_t\alpha_t\right) = \psi_t^*\left(\mathrm{d}\iota_{V_t}\alpha_t + \iota_{V_t}\mathrm{d}\alpha_t + \partial_t\alpha_t\right).$$

Proof. Fix t and define $\eta_h = \psi_{t+h} \circ \psi_t^{-1}$ on the current image, shrinking neighbourhoods and the range of h if necessary. Its initial value is the identity and $\partial_h \eta_h|_0 = V_t$. Joint smoothness gives the coordinate expansions

$$\eta_h(q) = q + hV_t(q) + O(h^2), \qquad D\eta_h(q) = I + hDV_t|_q + O(h^2).$$

These agree to first order with the autonomous flow of the fixed field V_t. The coordinate pullback formula uses just these first derivatives in h, so $\partial_h(\eta_h^*\alpha_t)|_0 = L_{V_t}\alpha_t$. Now $\psi_{t+h}^*\alpha_{t+h} =$

$\psi_t^* \eta_h^* \alpha_{t+h}$. The family $\eta_h^* \alpha_{t+k}$ is jointly smooth in h and k. The chain rule along $h = k$ adds its two partial derivatives at zero: $L_{V_t} \alpha_t$ and $\partial_t \alpha_t$. The fixed pullback ψ_t^* carries their sum back to the starting point. Cartan's formula gives the second equality. No autonomous group law for ψ_t was used. ∎

For example, $\psi_t(x, y) = (x + t^2 y, y)$ is generated from time zero by $V_t = 2ty\partial_x$. For $\alpha_t = tx\mathrm{d}y$ its pullback is $t(x + t^2 y)\mathrm{d}y$. The derivative is $(x + 3t^2 y)\mathrm{d}y$. Here $L_{V_t} \alpha_t = 2t^2 y\mathrm{d}y$ and $\partial_t \alpha_t = x\mathrm{d}y$. Pulling their sum back gives precisely that same derivative. Omitting the explicit time derivative would give a different answer.

For a fluid in Euclidean space, let v_t be its smooth velocity field.

Physical input. In the constant-density incompressible-fluid model, the velocity satisfies $\sum_i \partial_i v_t^i = 0$ at every point and time.

With Euclidean volume $\mu = \mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^m$, Cartan's formula gives $L_{v_t} \mu = \mathrm{d}\iota_{v_t} \mu$. Its coefficient is $\sum_i \partial_i v_t^i$, because only the derivative in the omitted coordinate survives. Thus this Lie derivative is zero. The time-dependent pullback formula, with $\partial_t \mu = 0$, therefore proves $\psi_t^* \mu = \mu$ wherever fluid trajectories exist. Every bounded parcel keeps its Euclidean volume in physical space throughout any interval on which its trajectories exist. Hamiltonian volume preservation concerns initial conditions in phase space.

Self-check. For the evolution $\psi_t(x, y) = (x + t^2 y, y)$ and the form $\alpha_t = (x^2 + ty)\mathrm{d}y$, compute $\frac{\mathrm{d}}{\mathrm{d}t} \psi_t^* \alpha_t$. Evaluate its $\mathrm{d}y$ coefficient at $t = 1$ and $(x, y) = (2, -1)$.

Answer. The velocity is $V_t = 2ty\partial_x$, so $L_{V_t} \alpha_t = 4txy\mathrm{d}y$ and $\partial_t \alpha_t = y\mathrm{d}y$. Pulling their sum back gives

$$\frac{\mathrm{d}}{\mathrm{d}t} \psi_t^* \alpha_t = \big(4ty(x + t^2 y) + y\big)\mathrm{d}y.$$

The requested coefficient is $4(-1)(2 - 1) - 1 = -5$. The term $y\mathrm{d}y$ comes from the explicit time dependence.

11.5 Volume forms and divergence

Definition 11.13. On an m-manifold, a *volume form* μ is a smooth m-form that is nonzero at every point.

Definition 11.14. For a smooth field X, the *divergence relative to* μ is the function $\mathrm{div}_\mu X$ satisfying

$$L_X \mu = (\mathrm{div}_\mu X)\mu.$$

Since the top exterior space is one-dimensional, this function exists uniquely. It is smooth: divide the local coefficient of $L_X \mu$ by the nowhere-zero coefficient of μ.

Definition 11.15. A flow *preserves* a form if its pullback equals that form on every time map's domain. Such a form is called *invariant* under the flow.

Proposition 11.16 (Divergence and volume preservation · *proved given smooth local flows*)**.** *In local coordinates, for* $\mu = \rho \mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^m$*, with* $\rho \neq 0$*,*

$$\operatorname{div}_\mu X = \frac{1}{\rho} \sum_{i=1}^{m} \partial_i(\rho X^i).$$

The local flow of X *preserves* μ *if and only if* $\operatorname{div}_\mu X = 0$ *everywhere.*

Proof. Since there are no forms of degree $m+1$, Cartan gives $L_X\mu = \mathrm{d}\iota_X\mu$. The contraction formula gives

$$\iota_X\mu = \sum_i (-1)^{i-1} \rho X^i \mathrm{d}x^1 \wedge \cdots \wedge \widehat{\mathrm{d}x^i} \wedge \cdots \wedge \mathrm{d}x^m.$$

After differentiation, only the x^i derivative of the ith coefficient survives. Moving $\mathrm{d}x^i$ to its ordered place gives another $(-1)^{i-1}$. Thus the resulting top-form coefficient is $\sum_i \partial_i(\rho X^i)$. Dividing by ρ proves the formula. If divergence vanishes, the finite-time pullback derivative is zero. Along each allowed time interval it is constant and equals μ at time zero. Conversely, differentiating invariance at zero gives $L_X\mu = 0$, hence zero divergence. ∎

At each starting point, Taylor expansion of the pullback gives

$$\varphi_t^*\mu = \big(1 + t \operatorname{div}_\mu X + O(t^2)\big)\mu.$$

Thus divergence is the instantaneous relative rate of volume expansion at that point. Positive divergence means expansion to first order.

For Euclidean volume, $\rho = 1$ gives the usual sum $\sum_i \partial_i X^i$. For example, $X = x\partial_x + y\partial_y$ has divergence 2 relative to $\mathrm{d}x \wedge \mathrm{d}y$. Its flow multiplies that form by e^{2t}. On the punctured plane choose instead $\mu' = (x^2+y^2)^{-1}\mathrm{d}x \wedge \mathrm{d}y$. The same field gives

$$\operatorname{div}_{\mu'} X = (x^2+y^2)\left(\partial_x \frac{x}{x^2+y^2} + \partial_y \frac{y}{x^2+y^2}\right) = 0.$$

The two derivatives sum to $2/r^2 - 2r^2/r^4 = 0$, where $r^2 = x^2+y^2$. Thus its flow preserves μ' while expanding Euclidean area. The choice of volume form matters. In polar coordinates, $\mathrm{d}x \wedge \mathrm{d}y = r\mathrm{d}r \wedge \mathrm{d}\theta$, so $\mu' = (\mathrm{d}r/r) \wedge \mathrm{d}\theta$. Dilation sends r to $e^t r$ and leaves $\mathrm{d}r/r$ unchanged. For an annulus with $r_2 > r_1 > 0$, its weighted area is

$$2\pi \int_{r_1}^{r_2} \frac{\mathrm{d}r}{r} = 2\pi \log(r_2/r_1).$$

Scaling both radii leaves this ratio unchanged, as the figure illustrates.

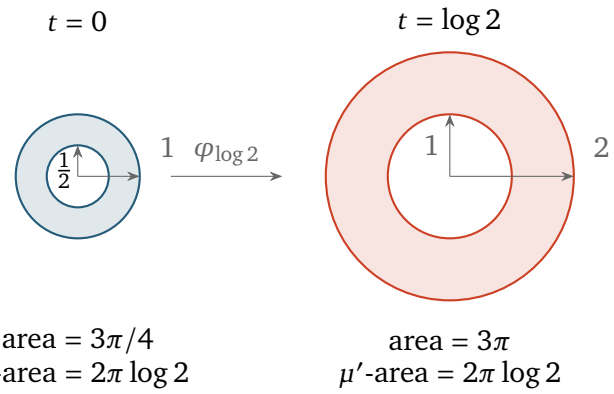


Figure 11.3: Dilation at time $t = \log 2$ sends the annulus with radii $1/2, 1$ to the annulus with radii $1, 2$. The weighted area for μ' stays fixed under this fourfold increase in Euclidean area.

11.6 Symplectic forms and Liouville's theorem

The canonical 2-form in §11.1 is closed and nondegenerate. Closedness will make its Lie derivative vanish. Nondegeneracy makes the Hamiltonian equation determine a unique field.

Definition 11.17. On a $2n$-manifold with $n \geq 1$, a *symplectic form* ω is a closed smooth 2-form that is *nondegenerate*: if $\omega_p(v, w) = 0$ for every $w \in T_pM$, then $v = 0$.

Definition 11.18. For a symplectic form ω on M and a smooth Hamiltonian function $H\colon M \to \mathbb{R}$, the *Hamiltonian field* X_H is determined by

$$\iota_{X_H}\omega = \mathrm{d}H.$$

Nondegeneracy makes $v \mapsto \omega_p(v, \cdot)$ an isomorphism. Its inverse has smooth matrix entries in charts, so the field exists uniquely and is smooth. We use the sign convention $\iota_{X_H}\omega = \mathrm{d}H$, which gives Hamilton's equations in the canonical coordinates above.

Lemma 11.19. *For a symplectic form on a $2n$-manifold, the top form ω^n is nowhere zero.*

Proof. Choose tangent vectors a, b with $\omega(a, b) = 1$. Every tangent vector v splits into a component $\omega(v, b)a - \omega(v, a)b$ in their plane and a component orthogonal to that plane for ω, meaning that its pairing with both a and b is zero. The restriction to the complementary subspace is nondegenerate, since a vector orthogonal to it and to the plane is orthogonal to the whole space. Repeat in the complement. In the resulting dual basis α^j, β^j, $\omega = \sum_j \alpha^j \wedge \beta^j$. Its nth power is $n!\alpha^1 \wedge \beta^1 \wedge \cdots \wedge \alpha^n \wedge \beta^n$, which is nonzero. Hence $\omega^n/n!$ is a volume form. ∎

For the canonical form the same wedge expansion gives

$$\frac{\omega^n}{n!} = \mathrm{d}q^1 \wedge \mathrm{d}p_1 \wedge \cdots \wedge \mathrm{d}q^n \wedge \mathrm{d}p_n.$$

The factor $n!$ counts the orders of the n distinct 2-form factors. Dividing by it gives the usual coordinate volume element.

Corollary 11.20 (Liouville · *proved given smooth local flows*)**.** *For a symplectic form and its Hamiltonian field,*

$$L_{X_H}\omega = 0, \qquad L_{X_H}(\omega^n) = 0.$$

Consequently its local flow preserves both the symplectic form and the volume form $\omega^n/n!$.

Proof. Cartan's formula and the defining Hamiltonian equation give $L_{X_H}\omega = \mathrm{d}^2 H + \iota_{X_H}\mathrm{d}\omega$. Both terms vanish because $\mathrm{d}^2 = 0$ and ω is closed. The derivation rule then gives zero on the n-fold wedge power ω^n. The finite-time pullback identity proves preservation. ∎

A set of initial positions and momenta can change shape under Hamiltonian motion, but its phase-space volume stays constant while the trajectories exist. This conclusion concerns the volume determined by $\omega^n/n!$. For example, the Hamiltonian $H = p^2/2$ on $\mathbb{R}^2$ gives $\dot{q} = p$ and $\dot{p} = 0$. Its flow is the shear

$$\varphi_t(q,p) = (q + tp, p), \qquad D\varphi_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \qquad \det D\varphi_t = 1.$$

The pullback is $(\mathrm{d}q + t\mathrm{d}p) \wedge \mathrm{d}p = \mathrm{d}q \wedge \mathrm{d}p$. The figure compares this changing shape with clockwise oscillator motion.

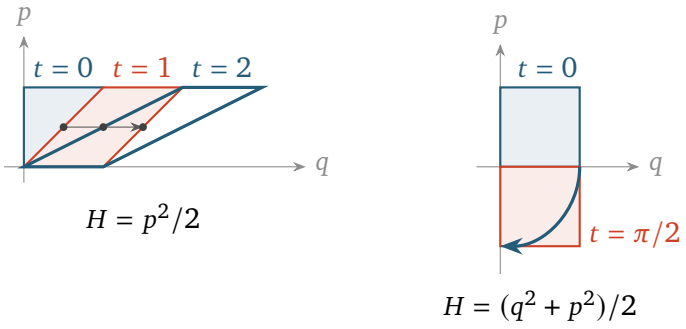


Figure 11.4: The shear sends a unit square to parallelograms of area one at times $0, 1, 2$, with dots tracking $(1/2, 1/2)$. The oscillator rotates the unit square clockwise through a quarter period and also preserves area.

11.6.1 The Poisson bracket and the field bracket

Definition 11.21. The canonical *Poisson bracket* of smooth functions on $T^*\mathbb{R}^n$ is

$$\{f,g\} = \sum_i \left(\frac{\partial f}{\partial q^i}\frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i}\frac{\partial g}{\partial q^i} \right).$$

The Hamiltonian field acts on a smooth function h by $X_H h = \{h, H\}$, as its coordinate coefficients show.

Example 11.22 (The bracket sign)**.** For one degree of freedom, take $f = q^2/2$ and $g = p^2/2$. Their fields are $X_f = -q\partial_p$ and $X_g = p\partial_q$. The coordinate bracket is $[X_f, X_g] = -q\partial_q + p\partial_p$. The Poisson bracket gives $\{f,g\} = qp$, whose field is $X_{qp} = q\partial_q - p\partial_p$. Thus this pair satisfies $[X_f, X_g] = -X_{\{f,g\}}$, using Lecture 9's bracket $[X,Y]h = X(Yh) - Y(Xh)$.

The minus sign holds for all smooth f and g. To prove it, differentiate the evaluation defining contraction under pullback. The product rule, with the tensor Lie derivative of Y, gives

$$L_X(\iota_Y\omega) = \iota_{[X,Y]}\omega + \iota_Y(L_X\omega).$$

Apply this identity with $X = X_f$ and $Y = X_g$. Liouville makes the last term zero. Since Lie differentiation commutes with d,

$$\iota_{[X_f,X_g]}\omega = L_{X_f}\mathrm{d}g = \mathrm{d}(X_f g) = -\mathrm{d}\{f,g\}.$$

Nondegeneracy therefore gives $[X_f, X_g] = -X_{\{f,g\}}$. This is a reversal of bracket sign, as for the linear fields $X_A(x) = Ax$ in Lecture 9. The Poisson bracket acts on functions, whereas the bracket of Lecture 9 acts on vector fields.

Self-check. State in one sentence what the Lie derivative of a form is and why we introduced it. Explain how Cartan's formula proves Hamiltonian volume preservation.

Answer. The Lie derivative of a form is the infinitesimal change of its flow pullback, introduced to test how a flow changes or preserves that form. For a Hamiltonian field, Cartan gives $L_{X_H}\omega = \mathrm{d}^2H + \iota_{X_H}\mathrm{d}\omega = 0$. The derivation rule gives zero on $\omega^n/n!$, so the flow preserves this volume form.

Summary. Contraction inserts a vector in the first slot and is a graded antiderivation. The Lie derivative of forms differentiates flow pullback. Cartan expresses it as $\mathrm{d}\iota_X + \iota_X\mathrm{d}$. A time-dependent form contributes its explicit time derivative as well. Zero divergence relative to a chosen volume form is equivalent to local flow preservation of that form. A Hamiltonian flow preserves its symplectic form and the associated top wedge volume. Lecture 12 integrates forms and relates their exterior derivatives to boundaries.

Reading. Tu, *An Introduction to Manifolds*, §20.3–§20.5, especially Theorem 20.10, pp. 229–230.